

Evaluating the network and usability characteristics of virtual reality conferencing

C Greenhalgh, A Bullock, J Tromp, and S Benford

This paper presents the results of the BT-funded 'Inhabiting the Web' project which conducted a series of twenty virtual meetings using the MASSIVE virtual reality teleconferencing system between five universities and BT Laboratories. The aims of the project were, firstly, to assess the characteristics of the network traffic generated by MASSIVE and, therefore, by extension, the network traffic generated by other peer-to-peer unicast-based collaborative virtual environments, and, secondly, to identify key human factors issues arising from the long-term and regular use of such a system. The paper gives an overview of the MASSIVE system and the organisation of the meetings and then presents an analysis of its results. The network analysis involves a statistical analysis of system logs in order to build a profile of typical user behaviour. This profile indicates how much users might be expected to engage in basic actions such as moving and speaking during a typical meeting and also to what extent these actions might be correlated among users. When combined with an understanding of MASSIVE's network protocols, this leads to a model to predict the network traffic generated by varying numbers of users. It also suggests new design possibilities such as 'group portals', caching of worlds and the use of multicast protocols. Usability reflections address the issues of embodiment, navigation and use of the spatial model of interaction as implemented in MASSIVE. New design possibilities are identified including higher level support for gestures on embodiments and so-called object-centred and context-sensitive styles of navigation.

1. Introduction

Recent years have seen a growing interest in the technology of collaborative virtual environments (CVEs) — distributed multi-user virtual reality systems that support mutual awareness and communication among their inhabitants [1]. The essence of CVEs is that multiple users share a computer-generated virtual world. Each user is able to freely navigate their own viewpoint within this world, is directly represented to other users through a process of embodiment, can communicate with other users employing a range of media (such as audio, text, graphics and video), and can directly manipulate objects within the world. There might be many potential applications of CVEs including virtual meeting rooms, shared 3-D information visualisations, battle and emergency simulations, and various kinds of entertainments.

BT's own research into CVEs has included the Virtuosi project, a three-year DTI/EPSCRC-funded project which was led by BT and which developed CVE demonstrators for the manufacturing and fashion industries [2], and The Mirror, a project between BT, Illuminations Television, the BBC and Sony, to stage a public trial of a 'dial-up' CVE alongside the television series *The Net*.

There are several potential benefits of this CVE technology [3].

- Support for natural, spatial communications skills

Humans have developed a range of subtle but important communications skills based on the use of space as a 'shared resource'. Key examples include the importance of gaze direction in conversational turn-taking and the deployment of a wide range of spatial gestures such as pointing, facing and many other more subtle accompaniments to speech. CVEs may allow users to employ these skills to an extent that is not possible with other communications technologies.

- Dynamic group formation

Many social situations, especially those involving more than a handful of participants, involve a degree of dynamic group formation — conversational groups come and go, groups split into sub-groups, sub-groups merge into larger groups and participants move between groups.

In contrast to previous teleconferencing systems which often assume a relatively static and predefined group membership, CVEs actively encourage 'mingling' behaviours and highly dynamic group formation.

- Scaling to larger numbers of participants

Following on from the previous observation, a common goal of CVEs is to support large numbers of simultaneous participants. CVEs support scalability in several ways — the perspective view offered by 3-D interfaces allows many participants to be seen at once, the use of attenuation with distance or even fully 3-D sound might help participants cope with many simultaneous audio streams, and the application of scoping techniques (such as ‘aura’, described in sections 2.1 and 5) allows users to limit their personal scope of communication in a densely populated environment. Current systems claim anything between ten and a hundred simultaneous participants, depending on the level of communication supported (e.g. audio or text), and other factors such as the level of user activity (indeed, factors affecting scalability are a key topic of this paper).

- Peripheral awareness

Computer-supported co-operative work (CSCW) research has indicated the importance of users establishing ‘peripheral awareness’ of the presence and activities of others [4]. By situating users within a 3-D space, CVEs implicitly support users in conducting locally focused communication, while at the same time ‘keeping an eye’ on more distant (i.e. peripheral) events. Some CVEs, such as the MASSIVE-1 system discussed in this paper, go further by introducing explicit mechanisms for establishing different levels of awareness within a shared virtual space.

- Partially overlapping media

An interesting property of human perception in the physical world is that we do not perceive the same objects at the same levels of detail in all of the available media. Typically, we can see some things that we cannot hear (e.g. distant objects) and hear some things that we cannot see (e.g. objects that are behind us). These ‘partial overlappings’ of media give rise to the potential for peripheral awareness. For example, we can listen to one person while keeping an eye on someone else who is approaching from the distance. Unlike other teleconferencing technologies which tend to depict the same objects across all media, CVEs naturally support the idea of trading off directed and peripheral awareness across different media.

- Representation of users and information within a single display space

Many teleconferencing technologies fail to adequately integrate communications channels with shared information. For example, shared drawing tools and editors are usually displayed in windows that are separate from the representations of the participants. In contrast, CVEs locate participants and information

within a single shared display space, thereby enabling participants to see who is attending to, or working with, what aspects of the information, from spatial cues such as body position, gaze direction and representations of virtual limbs.

However, to date there has been little formal experimentation aimed at verifying or refuting these various claims. There is also little understanding of the network traffic characteristics of CVEs, i.e. what kind of traffic will they generate or, conversely, what limits on scalability and complexity will be imposed by the use of different underlying networks (e.g. what could realistically be achieved using ISDN, ATM or the Internet). Answering these last questions will be a key step in provisioning any future CVE-oriented network services.

This paper presents the results of a programme of experimentation aimed at answering some of these questions. This programme has been carried out in the ‘Inhabiting the Web’ (ITW) project, a BT-funded project within the BT/JISC (Joint Information Systems Committee of the Higher Education Funding Council for England) Research Collaboration Programme. ITW ran from October 1995 until March 1997 and involved a collaboration between BT Laboratories and five UK universities (Nottingham, Lancaster, Manchester, Leeds and UCL). The project ran a series of network experiments of Nottingham’s MASSIVE-1 VR teleconferencing system [3] over the SuperJANET network in order to:

- construct a predictive model of network traffic valid for CVEs which are based on a peer-to-peer unicast network architecture — this involved the statistical analysis of system logs and network traffic measurements captured from virtual meetings,
- identify key social issues inherent in the use of CVEs, especially potential limitations of the technology or new emergent social phenomena — this involved a programme of ‘usability’ studies.

A secondary effect of the project has been to refine the methodologies required for dealing with these kinds of issues.

Section 2 of this paper provides a brief description of the MASSIVE-1 system and of the underlying spatial model of interaction upon which it is based. Section 3 describes the structure and organisation of the experiments. Sections 4 and 5 then focus on the network traffic analysis and model, while section 6 discusses the social issues that have been raised. Finally, section 7 brings us fully up to date by describing the design of the new MASSIVE-2 system and introducing two new projects, both involving BT and the University of Nottingham, which build on this work within the context of so-called ‘Inhabited Television’ applications.

2. A brief overview of MASSIVE-1

Firstly, a brief overview of the MASSIVE-1 VR-teleconferencing system is provided, describing both its functionality and its underlying network architecture. MASSIVE-1 was implemented as part of EPSRC-sponsored PhD research. The system runs on Silicon Graphics workstations over standard Internet protocols. A detailed description can be found in Greenhalgh and Benford [3].

2.1 Functionality of MASSIVE-1

MASSIVE-1 supports multiple virtual worlds connected via portals. Each world may be inhabited by many concurrent users, where each user can independently navigate their own viewpoint within the world. These users can communicate using combinations of graphics, audio and text media. The graphics medium renders objects visible in a 3-D space and allows users to navigate this space with six degrees of freedom. The audio medium supports both real-time conversation and the play-back of preprogrammed sounds. The text medium provides a plan view of the world via a window (or map) which looks down on to a 2-D plane across which users move (similar to some multi-user dungeons). Text users may interact by typing messages to one another or by emoting (e.g. smile, grimace, etc). These different media may be arbitrarily combined according to the capabilities of a user's terminal equipment. Thus, at one extreme, the user of a sophisticated graphics workstation may simultaneously run the graphics, audio and text clients (the latter providing a map facility and allowing interaction with non-audio users). At the other extreme, the user of a dumb terminal (e.g. a VT-100) may run the text client alone.

MASSIVE-1 is a specific VR-conferencing application and many of the features that one might expect of a general VR platform are not supported (e.g. full immersion and the ability to directly manipulate objects in a virtual world). The core functionality that is provided includes:

- moving with six degrees of freedom,
- communicating — speaking, triggering simple graphical gestures, and exchanging text messages,
- adopting a range of different camera views (e.g. out of body, over the shoulder, bird's-eye and mirror views) with the added ability to zoom the virtual camera in and out,
- posting text messages on simple message-board objects located within the virtual world.

Each user may specify their own graphics embodiment via a configuration file. In addition, some default embodiments are provided which are intended to convey the com-

munication capabilities of the users they represent (e.g. ears suggest audio capability, eyes suggest graphics capability, and so forth). A simple mouth object is also provided which appears whenever a user speaks above a specified volume threshold.

The most sophisticated aspect of MASSIVE-1 is the way in which all communications media are driven by the so-called 'spatial model of interaction' [5]. The spatial model defines a set of high-level mechanisms called 'aura', 'awareness', 'focus', 'nimbus' and 'adapters' which enable users to negotiate social interaction in terms of different levels of mutual awareness. These mechanisms also drive underlying network connectivity and level of detail for the different media.

The spatial model assumes the existence of some kind of shared space (i.e. a shared co-ordinate system), populated by objects which are exchanging information over a set of media types (e.g. graphics, audio, text and video).

As implemented in MASSIVE-1, aura is a bounding volume on a user's personal presence. Auras are medium specific, so that a user might have a different-sized aura for each medium. Consequently, different media have different scopes of interaction (e.g. a user might be seen at a greater distance than they are heard). A collision between two users' auras (as notified by the MASSIVE-1 aura collision detector process) results in a network connection being established. Figure 1 shows a space containing five objects which, as a result of their auras, are currently connected into two separate conversing groups for this particular medium (there might be other patterns of connectivity in other media).

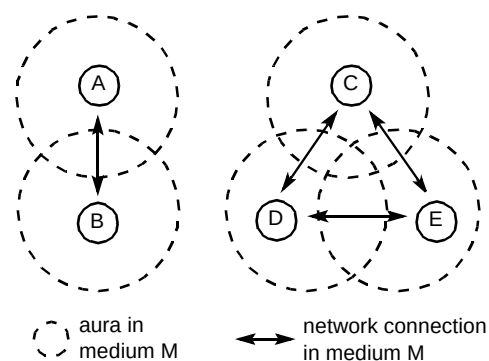


Fig 1 Auras used to establish network connections.

Once connections have been established in this way, the exchange of information across these connections is determined by mutual levels of awareness. Awareness in MASSIVE-1 is a quantifiable measure of the relevance that one object has to another at a given moment in time. Awareness can loosely be interpreted as an expression of desired quality of service for a given connection. Awareness is continually renegotiated as objects move around space

through the further mechanisms of focus and nimbus. Focus models an observer's allocation of attention across space and nimbus the corresponding projection of their presence into space. The more that an object is within your focus the more you become aware of it. The more an object is within your nimbus the more it becomes aware of you. As with aura, focus and nimbus can be defined on a per-medium basis.

The awareness that an observing object, *A*, has of an observed, *B*, is therefore a function of *A*'s focus on *B* and *B*'s nimbus on *A*. Figure 2 shows how different arrangements of focus and nimbus can result in different levels of awareness. The figure is only concerned with *A*'s awareness of *B* and hence only involves *A*'s focus and *B*'s nimbus. In Fig 2(a) *A* is focusing on *B* and *B* is nimbusing on *A*, resulting in *A* having full awareness of *B*. In Fig 2(c) the converse is true, so that *A* has no awareness of *B*. Figure 2 (b) shows the two cases where peripheral awareness occurs — in one of these *A* is focusing on *B* but *B* is not nimbusing on *A* (perhaps *A* is monitoring *B*?) — in the other *B* is nimbusing on *A* but *A* is not focusing on *B* (perhaps *B* is trying to interrupt *A*?).

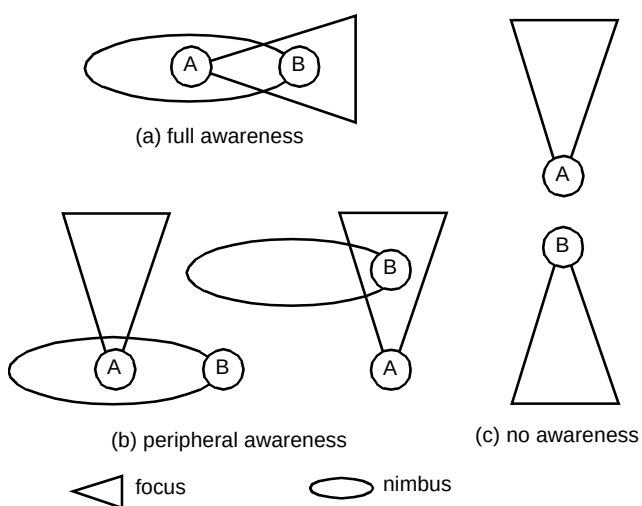


Fig 2 Focus, nimbus and awareness.

Users can manipulate aura, focus and nimbus, and hence awareness, implicitly by moving and re-orienting themselves within space or explicitly by choosing between different settings or even directly changing their shape and extent, e.g. retracting their aura in order to drop some network connections. Various adapter objects can also be used to dynamically alter aura, focus and nimbus. Examples of adapters in MASSIVE-1 include a podium which increases the range of aura and nimbus and a conference table which gives its users common auras, foci and nimbi, centred on the table.

MASSIVE-1 applies these concepts to each of its three media so that audio volume, graphical level of detail and the display of text messages are all awareness driven. For

example, audio awareness levels are mapped on to volume with the net effect that audio interaction is sensitive to both the relative distances and orientations of the objects involved. Unlike Fig 2 where focus and nimbus are simple containment volumes of space, MASSIVE-1 defines them as scalar functions across space. The shapes of these functions are such that focus and nimbus are at a peak directly in front of the user and then gradually drop off with distance from the user and with angle from their direction of face. Consequently, a range of different levels of awareness are possible.

In the current implementation, users can choose between three predefined settings for focus and nimbus — normal (intended for general conversation), narrow (intended for private conversation) and wide (intended for browsing). The wide setting is the default and is used most of the time.

Figure 3 shows how a virtual world appears in the graphics interface. Three embodiments can be seen, of which the one in the foreground ('Chris') is our own seen from an out-of-body camera view. A message-board can also be seen in the background. Figure 4 shows an example of the text interface. Top left of Fig 4 is a plan view of a virtual world; top-right is the key to the symbols used in this plan as well as an indication of current awareness levels between the user and objects in the world; and at the bottom the on-going text dialogue.



Fig 3 Three MASSIVE-1 embodiments.

2.2 Architecture of MASSIVE-1

This section focuses on the underlying network architecture of MASSIVE-1 as some knowledge of this is required in order to understand the analysis of network traffic later on (see Greenhalgh and Benford [3] for a detailed description). The processing model used in MASSIVE-1 is of independent computational processes communicating over typed peer-to-peer connections. Each computational process may support any number of interfaces, each of which is characterised by a combination of remote procedure calls (RPCs), attributes and streams in

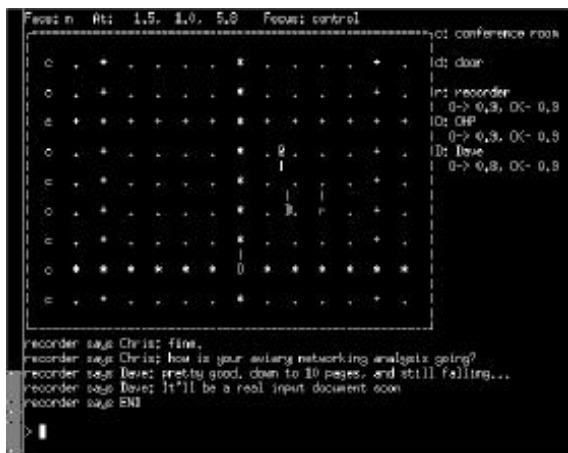


Fig 4 The text interface.

order to support a mixture of distributed VR functions (e.g. collision detection), distributed system functions (e.g. trading), and continuous media (e.g. audio).

In the interests of scalability, responsibility for detecting aura collisions and establishing connections is distributed. On entering a world, an object contacts the 'master trader' for this world and declares its interfaces (media). The master trader returns the addresses of the aura collision detectors responsible for these interfaces. At present there is a separate aura collision detector for each type of interface. These detectors monitor for aura collisions and, when they occur, pass peer interface references (combinations of IP address, port and a local interface ID) to the objects involved, allowing them to establish a connection.

Once connected, the calculation of mutual awareness levels using focus and nimbus is the responsibility of the peer objects. In the current implementation, objects are described solely by a sphere located in space, and focus and nimbus are described by scalar fields over space, which are sampled at the location of a peer object to generate focus and nimbus values. The current awareness function is multiplicative (i.e. focus and nimbus are multiplied to give awareness). This gives equal control to the observer and the observed, and is subtractive in nature, i.e. either party can force zero (no) awareness, but neither party can force awareness against the other's wishes. Current focus and nimbus functions describe conical volumes projecting forwards from an object with some additional transition volume where focus or nimbus drops off to a specified background level. The operation of focus and nimbus may therefore be controlled by a few key parameters such as conical angle, maximum radius, background level and transition rate, allowing variation from generally spherical to tightly focused regions.

Adapters exist in their own medium, complete with aura, focus and nimbus. As an object moves about, it may connect to an adapter as a result of an aura collision in the adapter medium. When the object's awareness of the adapter subsequently crosses some threshold the adapter is

triggered. The implementation of adapters exploits the ability to parameterise focus and nimbus functions; when triggered, an adapter passes some new parameters back to the object across the adapter medium. These new parameters replace the object's current parameters. Thus, an adapter may extend the range of focus and nimbus, may change their shape (i.e. conical angle), or may alter the way in which they fade to a background level. The object restores its normal parameters, when it moves away from the adapter.

3. An overview of the experiments

The 'Inhabiting the Web' project staged twenty virtual meetings using the MASSIVE-1 system. These meetings involved up to ten participants from BT Laboratories and the five universities listed above. On average, the meetings were held once a fortnight. A suite of meeting worlds and associated tasks were specifically designed for the project. The default workstation used by participants was a Silicon Graphics Indy, although a few participants had access to higher specification machines such as Indigo 2s. The meeting worlds were most often hosted on a twin processor Onyx 2 at Nottingham which was used for its computational power, not for its graphical rendering performance. The SuperJANET network provided connectivity between the partners, with appropriate tunnelling through the fire-wall at BT. Four kinds of data were captured from the meeting:

- direct video output from participants' machines at Nottingham and BT Laboratories,
- MASSIVE-1 log files were created at each site, where each file recorded and time-stamped all MASSIVE-1 events for a given local client (e.g. all movements, speech, gestures and so forth for an individual user),
- output from the 'tcpdump' command recording details of packet type (e.g. UDP, TCP), size and IP source, and destination addresses for packets which were observed on the local network (i.e. shared Ethernet) at Nottingham,
- completed questionnaires which were designed to capture information about users' backgrounds and experiences and which related to specific experimental tasks.

The twenty virtual meetings can be divided into four distinct phases — familiarisation and training, stabilisation, experimental tasks, and post-project social events. Familiarisation and training involved three initial meetings with standard MASSIVE-1 worlds, where participants were given some tutorial instruction on how to navigate and control their embodiments. They also involved some exploration of possible experimental tasks. Stabilisation focused on creating a stable experimental infrastructure in terms of the MASSIVE-1 software, virtual worlds, and also

user skills and procedures (e.g. data capture techniques). During this time, the Inhabiting the Web virtual worlds were created and refined and a number of problems with software and networking were identified. Technical problems included configuration of the firewall at BT, problems with variable network delays and setting appropriate time-out values in MASSIVE-1's protocols, problems with overly complex worlds and/or under-powered server machines, as well as more general bugs which had to be fixed.

The experimental phase of the project involved six meetings in which different tasks were carried out. The tasks were chosen to exercise a range of different MASSIVE-1 capabilities (e.g. movement, speech, interaction with the board and the resizing of focus) as well as different modes of co-operation (e.g. team games which required co-operation within teams and also competition between them). The content of these six meetings included:

- team and individual word games (e.g. teams pairing off categories of words which were written on two boards and writing the answers on another board) — this exercised navigation, conversation, dynamic group formation (e.g. to discuss answers in teams), board usage and peripheral awareness (e.g. overhearing the other team's discussions),
- searching for objects (e.g. fragments of text) or solving a sequence of clues located in different virtual worlds — this exercised the use of portals,
- dynamically altering audio focus so as to identify individual sound sources from among a large collection of sources — this exercised focus manipulation,
- the 'balloon debate' (a hypothetical discussion of which character should be thrown out of a rapidly descending balloon) — this exercised conversation among a relatively static and focused group,
- playing hide and seek in a virtual maze and completing a virtual obstacle course — these exercised navigation and awareness,
- disco dancing — this exercised rapid navigation and movement.

Finally, at the request of the participants (several of whom had never met face-to-face), several post-project meetings have been held. To date, the focus of these meetings has been the staging of a dramatic sketch (Monty Python's Fruit Sketch).

Figures 5—8 show different scenes from the ITW meetings and virtual worlds. Figure 5 shows several users gathering around the table in the central meeting space where Jolanda is giving a talk. Figure 6 shows several

participants interacting with three message-boards as part of a team game. Figure 7 shows the obstacle course located in the end-of-project party world. Figure 8 shows the final scene from the 'Fruit Sketch'.



Fig 5 Gathering around the meeting table.

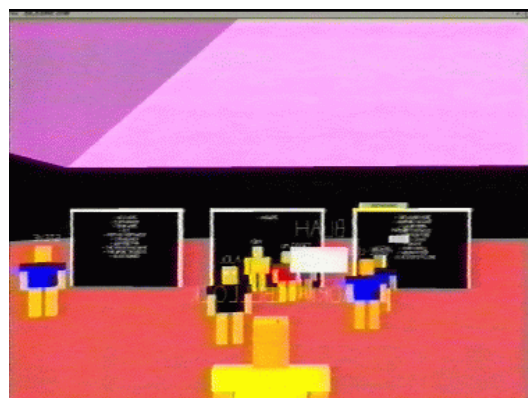


Fig 6 A word game on the message boards.

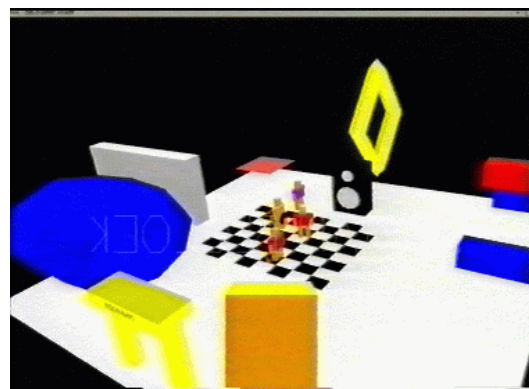


Fig 7 The obstacle course and disco world.

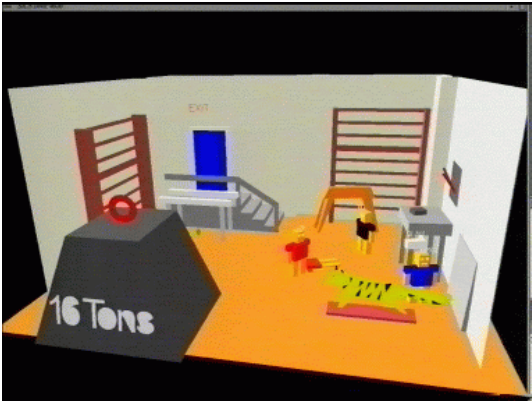


Fig 8 Enacting the 'Fruit Sketch'.

4. Network traffic analysis — user profiling

A primary goal of the ITW project was to explore the characteristics of the network traffic generated by MASSIVE-1 and, as a generalisation, by peer-to-peer unicast-based CVEs. More specifically, the aim was to be able to predict what volumes and patterns of traffic might be expected under different circumstances (e.g. for different numbers of users) and, conversely, what constraints on scaling would be imposed by the use of different network technologies.

The analysis proceeds in two stages. Firstly, a profile of typical user behaviour is developed which allows predictions of how an individual participant might be expected to behave (e.g. how much time they might spend moving, how often they might talk, and so forth) and also to what extent are different users actions correlated with one another (e.g. whether participants tend to move together in groups). Secondly, this profile is combined with knowledge of the MASSIVE-1 protocols in order to predict network traffic for different numbers of users. The first part of this analysis might be expected (within limits) to apply to a wide range of CVEs. The second part is specific to MASSIVE-1, although the general pattern that emerges might apply to the class of peer-to-peer unicast CVEs and a similar process could be repeated for other systems given sufficient knowledge of their protocols. The data presented in the following is based upon six experimental ITW meetings.

Quite early in MASSIVE-1's development, facilities were added to all programs, though especially the user client programs (the visual, audio and text clients), to generate time-stamped logs of key events. The events which can be recorded include:

- starting the application,
- moving to a new virtual world (through a portal),

- moving and changing orientation within a virtual world,
- updating the graphical view (i.e. rendering a single frame),
- changing focus and nimbus settings,
- making graphical gestures,
- speaking (or, more specifically, whether the user's graphical 'mouth' is displayed),
- sending and receiving network data.

Each program generates its own independent log file, and one of the problems of analysing this information is relating the information from different hosts, especially since each machine's system clock is typically different.

4.1 Analysis of movement data

The area of user behaviour which has proved most amenable to analysis based on the available MASSIVE-1 data is that of user movement. In particular the MASSIVE-1 event logs from a user client application record that user's moment-by-moment position and orientation within the virtual world, and also any change of world (due to passing through a portal). Firstly, an overview is provided of the movement data available, which is then used to address three issues for a subset of the meetings held using MASSIVE-1. The things that need to be established are:

- the fraction of time which people spend moving rather than stationary, and whether participants move simultaneously or independently,
- whether participants move through portals singly or in groups, and if, moving as groups, how long the transition takes,
- whether participants return to worlds which they have already visited, and if so after what period of time.

The implication of each of these is then considered:

- network and computational requirements for handling movement within virtual worlds,
- the scope for using multicast to handle inter-world transitions by a group's participants,
- whether 'world caching' (i.e. the user's local machine retaining a copy of the world for some time after they have left it in case they return again) would be effective and if so on what time-scale.

Figure 9 is a visualisation of movement-related information from the ITW meeting held on 25 September 1996.

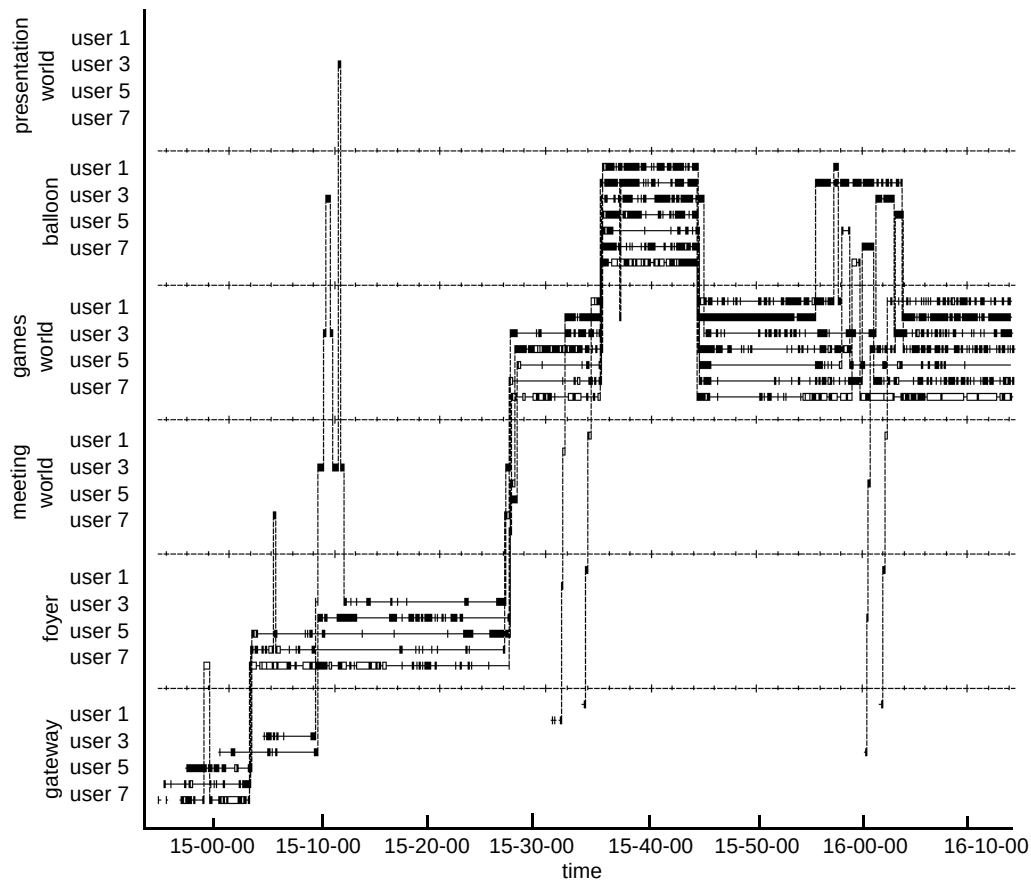


Fig 9 Visualisation of movement data from 25 September 1996 ITW meeting.

Time is shown along the horizontal axis, while a combination of participant and world is presented on the vertical axis. More specifically, the plot is divided into six horizontal bands which correspond to the six virtual worlds which were visited by participants during this meeting. The top-most band corresponds to the 'presentation world', the next to 'balloon world', and so on. Each participant is allocated a consistently placed slice within each of these bands; when a participant is present in a particular world this is represented by a horizontal bar in that participant's slice of the world band. When participants change worlds this is represented by vertical dashed lines between the worlds in question. Finally, the width of the horizontal line which represents a participant's presence within a world is varied according to whether that participant is moving or not; a thick line or box indicates that the user is moving, whereas a thin line indicates that they are stationary. So, considering the top-most band of Fig 9, it can be seen that participant number 3 jumps from 'meeting world' to 'presentation world' at approximately 15:10, moves about for a short time (probably checking if anyone else is there) and soon returns to 'meeting world'.

Time spent moving

Figure 10 shows the percentage of time which participants spent actively moving within a world. Each participant was considered independently, and for each visit

to a world the time for which they were moving and the total time for which they were present was established. In Fig 10 the horizontal axis corresponds to the percentage of time spent moving, while the vertical axis indicates the number of person-seconds for which that level of movement was observed. Both cumulative and point distributions are shown with 1% bucket sizes.

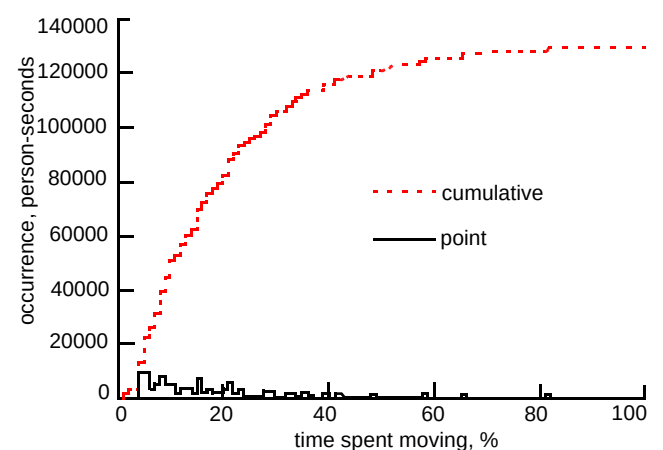


Fig 10 Percentage of time present in a world which was spent in motion.

The average percentage of time spent moving for all participants and worlds is 19.6%, though it is clear from

Fig 10 that this measure is highly variable. Independent of world visited, the figure for each participant varies from 7.2% to 28%. On the other hand, for each world, averaging over all of its visitors, the percentage of time spent moving varies from 7.5% (in the classical music world of the ITW end-of-project virtual party) to 54.6% (in the disco world, home of the disco dancing competition); in the main meeting world the average value was 16.7%.

The figures above are based on analysing each participant independently. In addition each world was analysed over the duration of the meetings to determine whether participants tended to move at the same time, or independently. Figure 11 shows the distribution of the number of participants in the same world at the same time who were moving. The solid line shows the observed distribution, while the dotted line shows the distribution that would be expected if all participants ignored each other and just moved when they felt like it (based on the same overall time spent moving, as previously determined). It is apparent from the graph that participants do co-ordinate their movements to some extent — the deviation from random activity is significant at the 99.9% level. However, the overall shape of the graph is very similar, and large numbers of participants moving simultaneously is possible — but less likely — in either case.

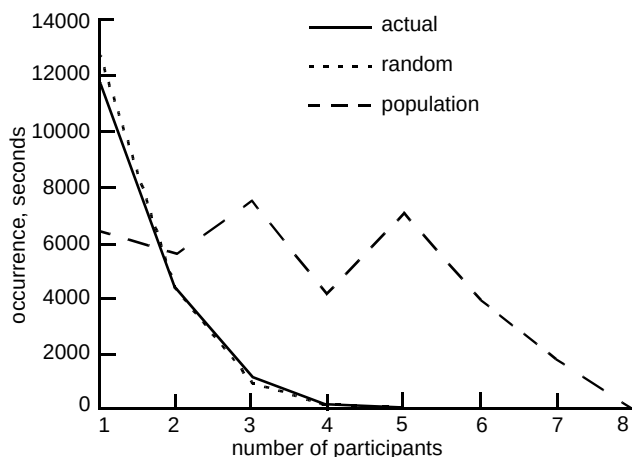


Fig 11 Distribution of numbers of virtually collocated participants moving simultaneously.

The time which participants spend moving and the amount of correlation between participants will have implications in terms of the network and computational requirements of communicating, processing and presenting each participant to the rest of the virtual world. Specifically, when a participant is moving, position and orientation updates need to be sent over the network and received and processed by each observing process. When dealing with non-deterministic agents (e.g. people) techniques such as dead reckoning may — or may not, depending on the application — reduce the number of such updates; however, they can never entirely eliminate the need for some of them.

Once a participant is stationary no further positional updates are necessary until they start to move again. Consequently the amount of time which participants spend moving can be a significant factor in assessing network and computer requirements.

The overall average percentage of time spent moving observed in the six meetings being analysed here was 19.6%, as noted above. This may be taken as a base-line value. However, there is also a large variation between individuals and also a significant task dependence, as indicated by the differing averages in different worlds. Consequently, the figure arrived at here should be treated as something of a rule of thumb, rather than a definitive answer. Additionally, when considering the combined instantaneous load due to several participants, it has been noted in this analysis that it is not valid to assume that their activities will be independent and uncorrelated. Rather, at least when participants are involved in a common task, there is a small but highly significant element of correlation, statistically, between their activities. This argues for additional caution when considering, for example, the scope for exploiting statistical multiplexing of movement-related traffic for larger numbers of users (i.e. being able to require or reserve less bandwidth on the basis that while some users are moving — and generating network updates — many other users will not be).

Group world transitions

Having discussed one key aspect of movement within a world, two aspects of moving between worlds are now considered. Firstly, the likelihood and form of co-ordinated inter-world transitions by groups of participants is considered, and, secondly, whether and in what circumstances participants return to previously visited worlds.

It would be expected from some of the activities organised in the meetings being analysed that group world transitions would occur. For example, participants would typically gather initially in the gateway world and wait for the others to arrive. Then at some point the meeting organiser would invite everyone to go through to the meeting world for the formal start of the meeting, and all of the participants would move — in a vaguely co-ordinated fashion — through to the meeting world. The purpose of this aspect of the analysis is therefore not simply to discover whether such transitions occur — it is already known that they do. Rather, it is the significance and character of such transitions that needs to be assessed.

For the purpose of the automated analysis, a group world transition is defined as an event in which two or more participants who are in a world at the same time move via a single portal-jump to another world so that they are together again. Figure 12 shows the incidence of singleton and group world transitions in the meetings analysed. The solid line

shows the number of incidents, while the dashed line shows the total number of participants involved in those incidents.

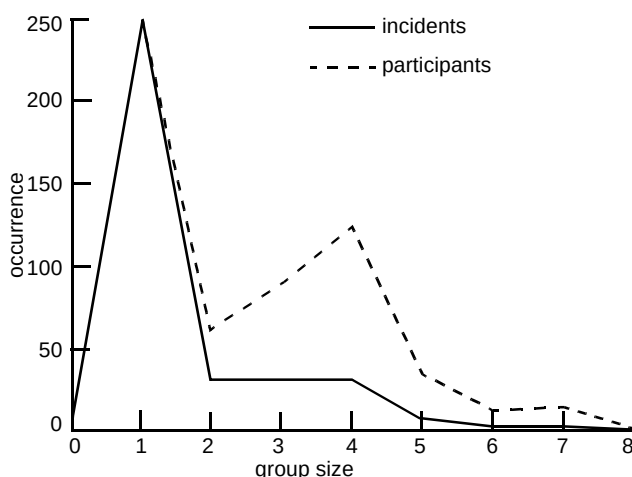


Fig 12 Distribution of group size for group world transitions.

Summarising the underlying data:

- participants jumped to new worlds on a total of 584 occasions, of which 337 (58%) were in groups of two or more,
- individuals or groups made world transitions on 350 occasions, of which 103 (29%) were group transitions and the average size of those groups was 3.27 participants.

Figure 13 shows the distribution of world entry delay for participants involved in group transitions, i.e. for each member of a group (excepting the leader) it shows how much time elapsed between the group leader and the group member reaching the destination world. The range of delay shown on the graph, up to 30 seconds, accounts for 203 (87%) of the 234 non-leading participants to make group world transitions; 104 (44%) of these occur within 5 seconds, while 159 (67%) occur within 10 seconds.

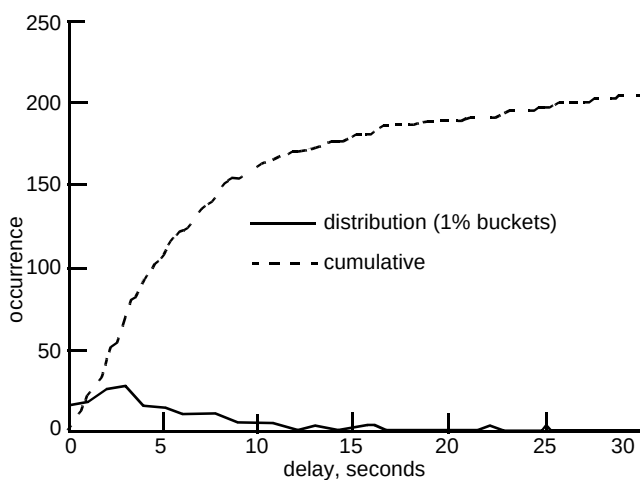


Fig 13 Distribution of group member arrival delay for group world transitions.

Group world transitions are important when considering the design and requirements of this kind of system, because moving to a new world is a significant event which will almost always involve an exchange of data between the participant's processes and the rest of the system. In MASSIVE-1, for example, on entering a new world the user's client applications terminate connections to objects in the old world, are informed of the existence and identity of nearby objects in the new world, establish network associations with each other, and exchange general and medium-specific information such as location, awareness, name, graphical appearance, etc. This can result in a significant but transient burst of network traffic and implied load on other processes.

Different systems will organise this information in different ways and obtain it from different sources, but there will still be some requirement for the participant's applications to learn about the new world. This makes the occurrence of group world transitions important in two respects:

- a co-ordinated movement by a large group could generate a much greater transient load than might be expected if inter-world movement were assumed to be independent and uncorrelated,
- specialised system support for group world transitions (for example, based on the use of network-supported multicasting of new world information to all group members) could not only alleviate this problem, but also reduce the total load relating to world transitions when compared to a model of independent and uncorrelated movement.

For example, for the meetings analysed, imposing a world transition delay of 10 seconds would both ensure that state transfers for the same world did not need to be performed more than once in any 10-second period, and would require approximately 325 unicast and 100 multicast state transfers rather than 584 unicast state transfers. Imposing longer delays on world transitions would increase the effectiveness of group transfers, by allowing more transitions to be effectively grouped, whereas shorter delays would include fewer transitions.

Before moving on to consider participants returning to worlds, a little must be said about the general applicability (or otherwise) of this result. As was noted at the beginning of this section, group world transitions were an organised aspect of the activities being analysed — will they occur in other applications and situations? Such a question cannot be answered definitively without gathering a great deal more data about a wide range of different applications and scenarios. However, some more subjective and tentative observations can be made:

- MASSIVE-1 is not alone in adopting a multiple world model with portals between worlds (see, for example, DIVE [6]),
- the world and portal model was widely accepted and effectively employed by participants.

It may be argued from these observations that a multi-world structure is a more generally useful and appropriate virtual design style. So it may be anticipated that group world transitions will occur in many applications involving formal or informal co-operation or interaction, e.g. as common interest groups form and dissolve, or as time-linked activities such as performances and meetings begin and end.

Returning to worlds

The final aspect of participant movement to be considered in this analysis concerns the incidence and character of return visits to virtual worlds, i.e. when a participant visits the same virtual world on more than one occasion in the same meeting.

Figure 14 shows the cumulative distribution of time elapsed between consecutive visits by individual participants to any world. There are a total of 353 return visits in the data analysed compared with 231 first visits. The average time lapse between leaving and re-entering the same world is just over 7 minutes, while half of these return visits occur within 3 minutes.

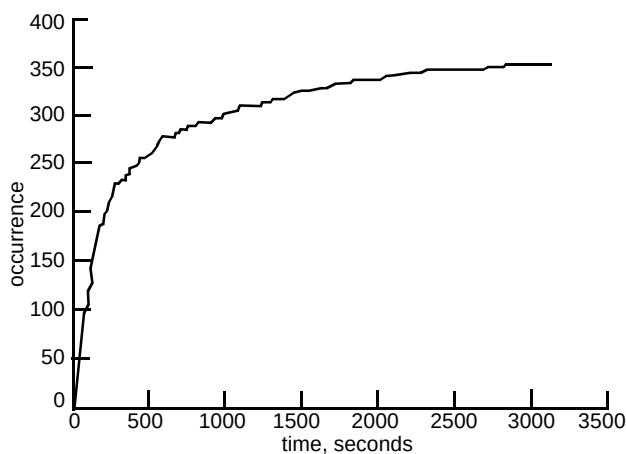


Fig 14 Distribution of time elapsed between consecutive visits to the same world.

The occurrence of revisiting worlds is important for the same fundamental reason as group world transitions were considered — because changing worlds is a significant event which implies an exchange of information with corresponding requirements for network and computing resources. In particular, world return visits are important because of the implications which they have for the

possibility and utility of world state caching in (or near) participant's applications. For example, for the data analysed, if each user client maintained a cache of world state for at least 6 minutes after visiting a world and if the worlds did not change on this time-scale then approximately 40% of new world state transfers could be satisfied from the cache.

The demonstration in the above example of the potential utility of world state caching must be qualified in two areas — the extent to which return rates can be generalised from the specific meetings analysed, and the extent to which worlds can be expected to remain static over the time periods in question and how this can be established by the applications.

Dealing with these issues in turn, it is clear that, if return visits were more common or sooner, caching would become increasingly effective, and conversely, if return visits were rarer or more delayed, caching would become less effective. As with group world transitions, it may be argued that most world return visits in the meetings considered were peculiar to the style and organisation of these meetings and not generalisable. However, there is again a (admittedly limited) counter-argument that these worlds and meetings were not established to demonstrate these effects, but that these effects emerge from natural choices of world and meeting structure which may also occur in other systems and applications. For example, the worlds for the ITW meetings were structured (for the most part) in a hierarchy, which is a very common method for organising related objects. A natural consequence of this is that the worlds closer to the root of the hierarchy were visited more frequently and were returned to more often, e.g. by participants in the process of moving from one task-oriented world to another. So it might be expected that revisiting of worlds will be a general effect.

Secondly, the potential validity of cached data — and establishing its validity or otherwise — is more clearly a general issue. In MASSIVE-1 worlds, the background content is static while the participants are highly dynamic. In other systems and applications the differences may be less clear cut. In any case, for caching to be effective there must be (significant) elements of world state which do not change between visits, and there must be some well-defined method of establishing what has not changed, and of efficiently combining cached and new information. This is left as an exercise to the reader.

4.2 Analysis of audio data

In addition to the analysis of movement and world transitions presented above, there has also been an analysis of audio-related activity in the same six MASSIVE-1 meetings. The figures that were available relating to participants' audio activity were records of audio data packets captured in the network traffic log, and events in the

MASSIVE-1 user client log files indicating when the visual 'mouth' was visible.

Each of these sources of data corresponds to a simple threshold test of audio volume. The visual mouth appears when the sound captured by the participant's microphone exceeds an experimentally chosen level, while the audio service sends audio packets (which are recorded in the network traffic log) when the sound captured by the microphone exceeds a lower experimentally chosen level. The aim here is to be generous in transmission (i.e. to transmit anything that might possibly be speech), but to only provide visual feedback via the 'mouth' when there is a high volume level so as to encourage people to speak up.

These sources of data will each be considered in turn before joint conclusions are drawn.

The data for logged network audio traffic from 'tcpdump' should include all audio in the same world as the principle meeting organiser. It extends to 33 198 person-seconds of audio from 125 571 person-seconds of apparent presence, i.e. the average speaking proportion is 26.44%. Apparent presence of audio participants is deduced from periodic audio timing packets which are sent by the audio service even when full audio data is not being transmitted. Figure 15 shows the number of simultaneous speakers for the data set (solid line) and the expectation if speaking was an independent event, with no correlation between virtually collocated participants (dotted line). The dashed line shows world population; if all the participants spoke continuously then this curve would result. The observed distribution differs from the uncorrelated case at the 99% significance level, but the difference at each data point is small — the difference is only significant because of the very large sample size.

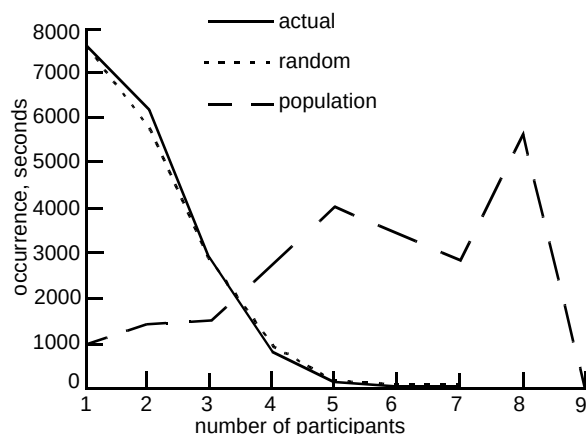


Fig 15 Correlation of network audio packets.

The data concerning the visual representation of a 'mouth' (actually more of an abstract speech balloon) is available where participants have correctly enabled MASSIVE-1's own logging facilities and have returned the event log files. The total participant time covered, 129 550

person-seconds, is similar to the network audio data. However, the amount of time for which the mouth is shown is much less — 9355 person seconds, or 7.22%. Figure 16 shows the number of simultaneous 'speakers' (actual and uncorrelated expectation) and world population for this data set.

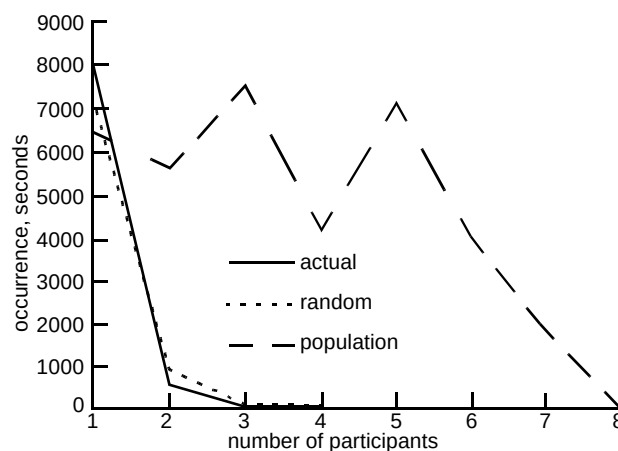


Fig 16 Correlation of visual 'mouth' data.

As for the network audio data the deviation of the observed number of simultaneous speakers from an uncorrelated distribution is significant at the 99% level, but small in percentage terms. Complete event logs for all participants are not generally available, and this is reflected in the shift of the world population distribution towards the left (fewer simultaneous participants) when compared to the network audio data. The distributions for speaking drop off much more steeply than for the network audio data partly because of this, but primarily because of the much smaller amount of speaking recorded.

Before the analysis, it was anticipated that the data for speaking might exhibit significant anti-correlation, reflecting the assumption that one participant would be less likely to speak when another participant was already speaking. In fact this was seen to a limited extent in the visual 'mouth' data, but not at all in the network audio data. At first sight this might suggest that the participants are ignoring one another, or that the network delays might cause a break-down in normal conversational turn-taking. However, a subsequent comparison between segments of the logged data and the video recordings revealed a more mundane reason — the majority of the apparent 'speaking' was actually due to background noise (e.g. typing, other activities in the office, breathing noise) and feedback from open speakers. This was especially true for the more sensitive network audio data. A number of observations and reflections can be made:

- use of open speakers (at least without sophisticated echo-cancellation techniques) can increase the apparent number of speakers and the resulting network load by replaying sounds back into the system,

- using more aggressive silence detection algorithms could reduce network traffic (moving the 26% speaking rate for network data towards the 7% speaking rate for the ‘mouth’), but at the risk of missing significant quiet utterances and non-speech noises (e.g. hmms, grunts and ahs),
- whatever the participant’s ‘real’ behaviour in terms of speaking (as determined by a human expert, for example), it is the behaviour that may be deduced automatically (as here) that is significant for assessing network and computational requirements.

Elements of the above analysis will be used in the next section which develops a network traffic model for MASSIVE-1.

5. A predictive model of network traffic

This section describes the development of a model of the network traffic required to support a MASSIVE-1 teleconference. A key element of this analysis is consideration for the way in which requirements change with the number of users. MASSIVE-1 has not been used for very large numbers of simultaneous users — the largest meeting to date involved ten participants. However, this traffic model allows us to consider the effects of much larger scale usage. Firstly, there is a description of the key elements of the user model employed and the additional assumptions required, and then briefly the main factors contributing to expected network traffic; these are combined with the user model to give a simple approximation of expected total network traffic. The section concludes by reflecting on the resulting traffic characteristics.

The key elements of the model and additional assumptions required are listed below:

- all users have a full complement of text, graphical and audio clients and use standard embodiments,
- the most significant user events are moving and speaking (text messages and gestures are ignored) — for these, users move 20% of the time (section 4.1) and speak (or rather send network audio data) 25% of the time (section 4.2),
- the average frame rate is 6 Hz (observed in use on a Sun 10ZX, and adequate for normal use),
- users move between worlds or groups of users such that they change the peers with which they interact once per minute (this value is somewhat higher than might be the case in the experiment meetings, but it will be seen that the related traffic component remains a relatively insignificant part of the total),
- all users interact with (on average) M other users, e.g. users might organise themselves into variable and changing groups of average size $M+1$ participants,

- the contribution of background objects is much less than that of users, and can be ignored (in MASSIVE-1 a background object generates approximately one tenth the traffic of a participant).

In MASSIVE-1 the main causes of network traffic are co-ordinating multiple user clients, updating the aura collision manager (due to movement, world transition or aura adaptation), establishing new associations with other users and objects upon aura collision, and interacting with other users and objects while in aura range. Of these four items, the first two are independent of the number of other participants using the system; the third — establishing new associations — depends on the rate at which groups or associates change; the fourth — interaction — depends on the number of other users and objects within aura range at any time (denoted by M).

The three key events considered are user movement, speech, and the arrival of a new interaction partner (i.e. an aura collision and subsequent data exchange). Table 1 shows the basic traffic generated by each of these events independent of any particular assumptions about user behaviour; each has a component which is per user only (the upper row), and a component which is also dependent on the number of peers which each participant has (the lower row). This is combined with the user model (average movement rate of 1.2 Hz, speaking rate of 25%, peer change rate of once in 60 seconds) to give the average resulting network traffic bandwidths in Table 2.

Table 1 Network traffic resulting from key events.

	Movement (kbyte/step)	Audio (kbyte/sec)	New peer (kbyte/peer)
Per user	1.2	0	2.1
Per peer per user	2.1	8.3	13.2

Table 2 Average network bandwidths.

	Movement (kbyte/sec)	Audio (kbyte/sec)	New peer (kbyte/sec)	Total (kbyte/sec)
Per user	1.4	0	<0.1	1.4
Per user per peer	2.5	2.1	0.2	4.8

Traffic model

From the final column of Table 2 an overall expression is obtained for the total expected network bandwidth (in kbyte/sec) as a function of the total number of participants, N , and the average number of other participants in aura range, M :

$$B = N (4.8 M + 1.4) \quad \text{..... (1)} \quad 113$$

For constant group size (i.e. constant M) this will be proportional to N , the total number of participants. On the other hand, if all users are in aura range (i.e. effectively a single group) then $M = N - 1$ and the maximum possible bandwidth is obtained:

$$B_{max} = 4.8 N^2 - 3.4 N \quad \dots (2)$$

This is equivalent to not making use of aura or multiple worlds to structure and restrict interaction. Figure 17 shows the resulting total bandwidth against number of simultaneous participants for different group sizes (different values of M).

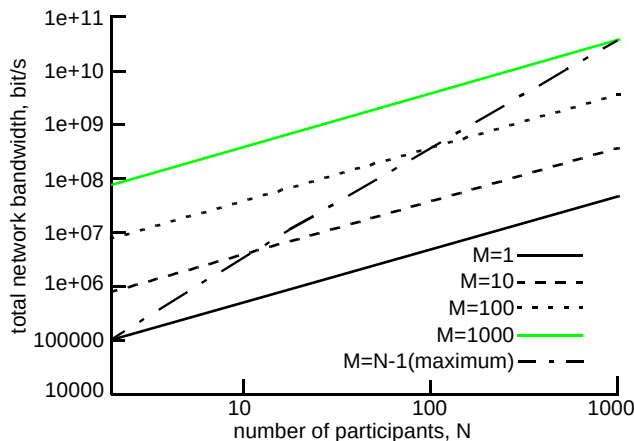


Fig 17 Total network bandwidth against number of participants for a range of group sizes.

Figure 18 shows the trade-offs between total number of participants and average group size for a range of total network bandwidths.

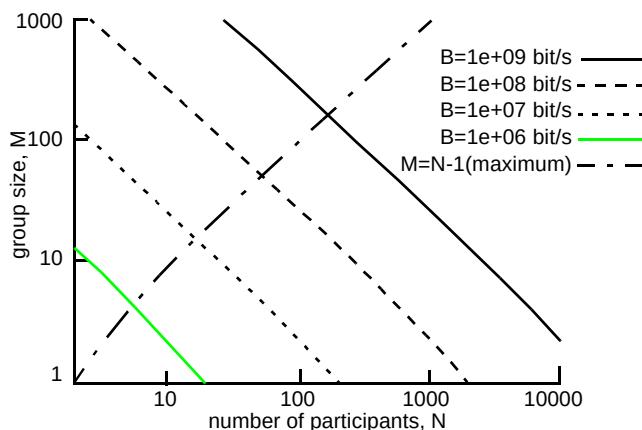


Fig 18 Trade-off between number of participants and group size for a range of total network bandwidths.

The assumptions that underlie this simple model of total network traffic have been made explicit, and, of course, they limit the accuracy or appropriateness of the model in other situations or scenarios and should be borne in mind

when applying these results. The main observations concerning the scope of applicability of the model are listed below.

- Different levels of user activity will result in different numbers; however, the overall form of the results will be the same.
- Different systems will use different protocols, giving rise again to different exact numbers. In particular, MASSIVE-1 may be seen to be rather inefficient (e.g. duplicating movement messages for each medium). However, it is relatively uniform in its inefficiency, so that again the same kind of behaviour would be observed, albeit for slightly greater numbers of participants or group sizes.
- As noted, MASSIVE-1 uses unicast-based peer-to-peer networking; the model and inferences are specific to this approach.
- The impact of background objects has been ignored. This may be appropriate in a teleconferencing application, but would not be appropriate in a database visualisation with thousands or millions of non-user objects. To apply the model in this kind of situation these objects would need to be taken into account.
- A certain rate of change-over of peers and of peer complexity (e.g. geometry size) has been assumed. With these assumptions the contribution to total traffic of establishing new peer associations is very small (less than 5%). However, a combination of rapid change-over and much greater peer complexity could dramatically increase this traffic component and the total requirement.

Nonetheless, the model (and MASSIVE-1 itself) remains representative in form, if not fine detail, of any CVE system architecture based on unicast peer-to-peer communications in conjunction with spatial trading.

Use of aura

The model and expected network traffic demonstrate the potential of auras to reduce network requirements and/or increase the potential number of participants when compared to a system without auras. Most significantly, the use of auras reduces the total network bandwidth from $O(N^2)$ (equation 2) to $O(NM)$ (equation 1). The management traffic overhead for interacting with the trader is not large (it is one component of the per user traffic in Tables 1 and 2) and is only $O(N)$. Using auras to scope (and limit) interaction in this way has the same effect as enforcing a reasonably even distribution of participants between different worlds. However, auras and spatial trading have significant advantages over the simplistic approach of

splitting participants according to world in that they naturally include division by world as a special case, they allow gradual, natural and visible transitions between groups within the same world, as opposed to sudden and discrete jumps between worlds, they avoid the need for invasive system intervention such as barring access to busy worlds, and they support flexible control and graceful degradation through interactive (and potentially automatic) modification of aura, e.g. pulling in one's aura to reduce system and network load in a busy area.

Scalability of unicast-based CVEs

Even with spatial trading to manage the number of active peers, it is clear from this traffic model that peer-to-peer interaction is the dominant component of network bandwidth, even with relatively small numbers of active peers (this is the per-peer, per-user component in the model, which is $O(NM)$). In a CVE there are elements of interaction which are specific to a single pair of peers; this is especially the case in a CVE based on mutually negotiated awareness as MASSIVE-1 is. However, the normal concept underlying a CVE is of a shared and primarily objective virtual world — participants can agree about the locations and properties of objects and each other. This objectivity and agreement are significant for enabling co-operation. Consequently, much if not all of the information being communicated between participants' processes is objective in character, i.e. observer independent.

The unicast-based networking employed in MASSIVE-1 restricts each network packet (each message or block of information) to be delivered to a single destination. Consequently, to send the same information to a number of destinations (such as a number of peers) requires one packet to be sent to each destination. These packets will normally travel over the same network links and segments for some or all of their journeys. It is this duplication of packets which gives rise to the $O(NM)$ limiting term in the network traffic model.

However, network multicasting allows the same packet to be delivered to potentially many recipients. With shared physical medium networking technologies such as Ethernet and FDDI, the same packet can be directly received by any number of machines on that network. The addition of multicast routers allows a single packet to be distributed to many recipients over wide-area and inter-networks. Because the multicast routers defer duplication of packets until the last possible moment (i.e. when the network divides), duplicate packets can be avoided on all network links and segments [7].

Ideal multicasting could reduce the total network bandwidth (in kbyte/s) to:

$$B = N (0.2 M + 6.0) \quad \dots (3)$$

This is shown in Fig 19, which is directly comparable to Fig 17 for unicast-based networking.

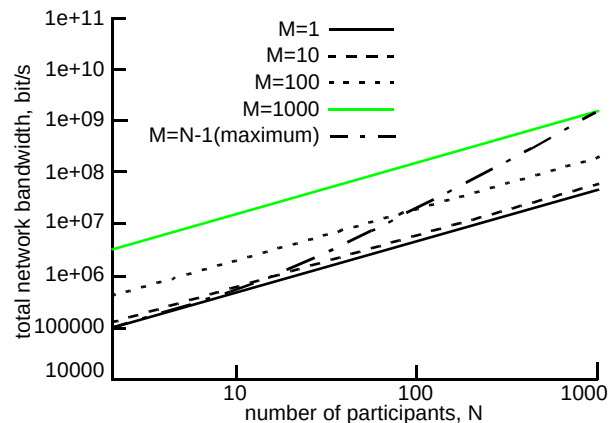


Fig 19 Total network bandwidth against number of participants for a range of group sizes assuming ideal multicasting.

All updates (movement and audio) are assumed to be ideally multicast. The remaining much-reduced $O(NM)$ term corresponds to forming the initial association with a new peer and exchanging information, such as appearance. There are a number of optimistic assumptions in this new model which would tend, in reality, to increase the $O(NM)$ term:

- this ignores the possibility of specifically peer-to-peer elements of information exchange which is inherently $O(NM)$ (this might include aspects of awareness negotiation, for example),
- it also ignores any multicast management overhead associated with joining or leaving groups,
- it assumes reliable multicasting — for at least some updates, though probably not audio — with no overhead or per-receiver growth in traffic,
- it assumes (as does the model already presented) that total network traffic is the most important factor — this assumption would seem to be more realistic for the case of unicast network architectures,
- it depends strongly on the rate at which new peer associations are formed, which will be highly application and context dependent.

Nevertheless, the potential improvement is dramatic — for an average group size of ten it gives a six-fold reduction in required bandwidth (or equivalently six times as many participants in the same bandwidth); for larger group sizes the benefits are greater still, up to a factor of more than 20 times. So for an objective CVE the potential scalability of a unicast-based system will typically be much less than for a

multicast-based system, as far as total network bandwidth is concerned. This observation has been a key factor in motivating the development of MASSIVE-2 as discussed at the end of this paper (section 7).

6. Usability

A brief summary of the usability issues raised by the ITW experiments is now presented. A replicated time-series design was followed for the usability analysis, in which the repetitive collection of data involved constructs such as satisfaction, experience, ease of use, group involvement and awareness. In addition, each session was used to explore new constructs such as dealing with multiple virtual embodiments, distributed awareness and switching between the virtual environment and the real environment. Human behaviour and experiences were captured using questionnaires, and video-recordings were made of each experiment. Over one hundred questions were asked in total during the experiments. The participants were all researchers in the area of virtual environments and so had a high level of familiarity with the underlying concepts. Several of them had considerable prior experience with MASSIVE-1 but a few had never seen the system prior to the project.

The questions were collected through the World Wide Web, and consisted of attitude statements with Likert-scales, and open-ended questions. For example, after an experiment aimed at learning a feature of the interface called 'Focus', which allows a user to change their degree of awareness of each other and the environment: 'How easy or difficult do you find it to use the focus commands? Please score on a scale of 1 to 7, if 1 = very difficult, and 7 = very easy,' and 'Please explain how you would change the commands for the focus feature to improve its ease of use.' Interviews were used as a follow-up for some of the more interesting observations. The remainder of this section focuses on three of the key issues that emerged from this work — user embodiment, navigation, and the use of the spatial model — drawing out some implications for future CVE system design.

6.1 User embodiment

Many of the findings come down to the suggestions for improvements of the degree of control of users' embodiments, of missing non-verbal interaction mechanisms, and of the lack of natural mappings to embodiment controls. The need for improved embodiments expressed by the subjects in the ITW study suggest that automatic embodiment behaviours ought to be introduced, in order to allow higher level interaction abilities. For the purpose of this paper, the suggested improvements can be summarised as follows:

- supporting automated actions, e.g. gestures for waving, nodding, running, walking and facial expressions, and also support for higher level automated actions, such as the actions a chairperson would want to use to open a meeting,
- objects should react to the user, instead of the user to the object in order to interact with it, e.g. having the object's functions triggered and made available by approaching the object,
- buttons for often-used behaviours, e.g. having an 'applaud' button which produces the sound of hand clapping — the more users press the button at the same time, the louder the applause,
- transparency of control, e.g. it should be obvious who is the current speaker, or who is currently manipulating an object, and a user should be able to go against the automatic behaviour of an object when appropriate,
- passive capture of often-used behaviours, e.g. in a virtual meeting users want to take their usual place around the table, and adjust their view to be able to see all other participants around the table.

The changes to MASSIVE-1 which are needed to satisfy these demands are best addressed at the level of the user embodiment. By providing the user with a programmable interface to the embodiment, the users should be able to create their own automatic behaviours, control their embodiment's reactions to objects, and create sets of automatic actions which they feel are appropriate for certain recurrent tasks. Users should also be able to share such sets of behaviours with each other. This creative editing and inheriting of automatic actions should also contribute to a naturally evolving CVE on a social and functional level.

Another more radical option would be to introduce some kind of immersive or semi-immersive interface, perhaps through the use of a head-mounted display for the former or through the use of magnetic trackers to capture users' gestures in front of a large projected interface for the latter. This possibility was mentioned repeatedly by several users, and could make an interesting contrast to the semi-automated approach to embodiment control.

6.2 Navigation

Moving about the virtual environment provided many users with a significant challenge. In particular, difficulties with fine-grained movements, such as aligning to other people, tables, boards and other objects, proved to be a major source of frustration. Several design options might be explored in order to address these problems.

- Automatic alignment with designated objects such as meeting tables or other users — sophisticated variants of this strategy might also align the user with other users who were themselves aligned with the object. For example, on approaching a meeting table one would be steered towards a vacant position that afforded the best possible view of the other participants. Such alignment might also operate dynamically as in the example of being automatically aligned to face the current speaker (although such ideas would need careful implementation).
- Automatic tracking of moving objects as a further extension to automatic alignment.
- Object-centred navigation, as a logical extension of the above, where a user's movements would all be relative to a designated focal point. For example, after designating a new focus, one would be able to sweep around it and zoom in and out on it without ever losing it from sight.

As a further note, different tasks might require different forms of navigation. For example, movements around a meeting table might suit an object-centred style, whereas navigation of an open space might suit a more traditional Cartesian navigation style. This leads to a final design proposal of context-sensitive navigation, where different styles of navigation (and indeed other forms of interaction), would be triggered within different contexts. For example, on approaching a meeting table, one might automatically swap from Cartesian to object-centred modes of navigation.

6.3 *Visibility and malleability of spatial model mechanisms*

The final issue arises from the use of the spatial model concepts of focus and nimbus in interacting with the board object and in the task where users have to selectively filter a collection of audio sources by moving about a world with a narrow audio focus setting. Users found it difficult to reason about the effects of focus and nimbus or about awareness relationships with other users and objects. For example, it was difficult to tell whether a message board had been successfully engaged or whether another user was aware of oneself. Although this information was available in the text interface and a range of basic settings were provided for focus and nimbus, it is proposed that, where such mechanisms are used, it is important that they are made sufficiently visible and easily malleable, e.g. other users' embodiments might depict the level of their awareness of the observer for a range of different media.

7. **Conclusions and recent developments**

This paper has presented the results of a series of experiments with the MASSIVE-1 collaborative virtual environment running over the SuperJANET network between

BT Laboratories and five UK universities. The main objective of the experiments was to gain an understanding of the kinds of network traffic generated by MASSIVE-1 and, by extension, by the class of CVEs based on a unicast peer-to-peer network architecture. A secondary objective was to explore some of the usability issues associated with this technology.

The consideration of network traffic issues involved an analysis of two kinds of data — multiple system log files generated by different user clients and output from the 'tcpdump' network monitoring command — and proceeded in two stages. Firstly, a general profile of user activity was developed which provides data on the typical actions of users across a range of tasks (i.e. how often do people move, talk, change worlds, etc) and importantly, the extent to which these different actions are correlated. Secondly, the user profile is combined with information about MASSIVE-1's network protocols in order to develop a model of predicted network traffic for different numbers of users in different sizes of mutual awareness groupings. Key findings for this work were that:

- on average, users move about 20% of the time and speak (i.e. send network audio data) about 25% of the time, although these figures vary with task,
- when participants are involved in a common task, there is a small but, statistically, highly significant element of correlation between their activities — this implies that some caution may be needed when considering the use of techniques such as statistical multiplexing of the associated network traffic,
- both world transitions and the frequency and timings of return visits to worlds are important events as they might be expected to generate considerable network traffic, group transitions (i.e. two or more participants using a portal within a short space of time) being a common occurrence as was returning to previously visited worlds — given that these results generalise to other applications, this has important implications for the design of 'group portals' which, at a cost of small delays to individuals, enforce group transitions which can be efficiently dealt with at the network level (e.g. by multicasting world state information) and also for the use of caching,
- the use of more aggressive silence detection algorithms could help reduce audio traffic, although at the risk of missing potentially significant non-speech and background noises.

The predictive model of network traffic for MASSIVE-1 has led to two main findings. Firstly, aura is a useful concept for limiting network traffic in CVEs. For an application involving N simultaneous users, auras would reduce the total traffic from being $O(N^2)$ to being $O(NM)$, where M is the

number of users in a mutually aware group of intersecting auras. Secondly, the use of network multicasting would be a significant factor in further scalability. For example, given an average group size of ten participants, there could be a six-fold reduction in required bandwidth rising to a factor of more than twenty times for larger groups.

Although specific to MASSIVE-1 and to the kinds of task undertaken in the experiments, it is believed that these findings will be broadly consistent with the behaviour of unicast peer-to-peer CVEs used for a range of different activities, and also that this general method of analysis should be of interest to those engaged in analysing other classes of CVE or even other kinds of co-operative system.

The exploration of usability issues mainly involved the analysis of questionnaires. Key issues raised by the participants include the need for more expressive and controllable embodiments, problems with navigation (especially managing 'fine-grained' alignment to objects and other participants), and the need to make spatial model concepts such as focus and nimbus more visible and malleable at the user interface. These findings suggest several new design possibilities for MASSIVE-1 including the association of more powerful animations with embodiments and also options for automatic alignment, object tracking, and object-centred and context-sensitive navigation.

In conclusion, two recent developments, which follow on from the ITW project, are mentioned here. The first of these is the development of the MASSIVE-2 system (also sometimes known as CVE) [8]. MASSIVE-2 represents a considerable extension of MASSIVE-1 both in terms of functionality and network architecture. Key extensions include:

- an extension to the spatial model of interaction-called third parties which support indirect awareness relationships between objects and which can be used to model a range of contextual and environmental effects on communications, with third party objects able to be used to model a diverse range of structures including:
 - nested regions whose boundaries have varying degrees of transparency,
 - mobile crowds which provide aggregate views of the inhabitants,
 - shared objects which manipulate the mutual awareness of their users,
 - group vehicles,
- the mapping of this extended model on to an underlying multicast architecture — essentially, the dynamic and nested structure of a world (e.g. nested regions and mobile crowds) is mapped on to a self-

configuring hierarchy of multicast groups, with protocols then being defined to manage group membership as a result of movement within a virtual world,

- support for direct interaction with the world and additional aids to navigation as well as the provision of a general application development environment.

As a final note, at the time of writing, Nottingham University and BT Laboratories are establishing a further collaboration in the form of the eRENA project within ESPRIT IV Long Term Research. eRENA will involve a collaboration between technologists, artists, broadcasters and social scientists to explore how large-scale CVEs such as MASSIVE-2 can be used to support new forms of cultural experience in areas such as art, entertainment, performance and so-called 'inhabited television' — the deployment of very large scale CVEs into the home as a new form of socially participative television.

References

- 1 Benford S D, Bowers J M, Fahlén L E, Mariani J and Rodden T R: 'Supporting co-operative work in virtual environments', *The Computer Journal*, **37**, No 8, Oxford University Press (1994).
- 2 Rogers A S: 'Virtuosi — virtual reality support for groupworking', *BT Technol J*, **12**, No 3, pp 81—89 (July 1994).
- 3 Greenhalgh C M and Benford S D: 'MASSIVE: A virtual reality system for teleconferencing', *ACM Transactions on Computer Human Interfaces (TOCHI)*, **2**, No 3, pp 239—261, ACM Press (September 1995).
- 4 Heath C and Luff P: 'Collaborative activity and technological design: task co-ordination in the London Underground control rooms', in Bannon L, Robinson M and Schmidt K (Eds): 'Proceedings ECSCW '91', Kluwer, Dordrecht (1991).
- 5 Benford S D and Fahlén L E: 'A spatial model of interaction in virtual environments', *Proc Third European Conference on Computer Supported Cooperative work (ECSCW'93)*, Milano, Italy (September 1993).
- 6 Fahlén L E, Brown C G, Stahl O and Carlsson C: 'A space based model for user interaction in shared synthetic environments', in *Proc Inter CHI'93*, Amsterdam, ACM Press (1993).
- 7 Deering S: 'Host extensions for IP multicasting', *IETF RFC 1112* (August 1989).
- 8 Benford S D, Greenhalgh C M and Lloyd D: 'Crowded collaborative virtual environments', *Proc ACM Conference on Human Factors in Computing Systems (CHI'97)*, pp 56—66, Atlanta, Georgia, US (March 1997).



Chris Greenhalgh is a lecturer in the Department of Computer Science at the University of Nottingham, where he is a member of the Communications Research Group. He is the creator of the MASSIVE-1 and MASSIVE-2 (also known as CVE) multi-user distributed virtual reality systems. He gained a first class BA in Electrical and Information Sciences from Cambridge University in 1991. He then worked in the data communications group at the GEC Hirst Research Centre for two years before moving to Nottingham to study for a PhD in large-scale collaborative virtual environments

which is nearing completion.



Jolanda Tromp gained her Masters degree in Psychology from the University of Amsterdam and a Bachelors in Art from the Rietveld Academy in Amsterdam. She has worked on the ESPRIT III-funded COMIC project and the BT-funded Inhabiting The Web project. She is currently working on the ACTS-funded COVEN project. Her research interests include methodologies for CVE research, taxonomies of CVEs, social conventions in CVEs and mixed realities. She is on the programme committee of CVE'98.



Adrian Bullock is a researcher in the Department of Computer Science at the University of Nottingham. His research interests include virtual reality, CSCW and spatial access control, and he is finishing a PhD in this last area. Previous projects have included the ESPRIT funded COMIC project and the BT/JISC-funded Inhabiting the Web project, and he is currently working on the ACTS-funded COVEN project.

He has extensive experience of networked virtual reality trials with a number of platforms.



Steve Benford is a Reader in Computer Science at the University of Nottingham. His research interests are centred on 'social computing environments' — the design of technologies to enable new forms of human communication. In recent years, he has focused on the development of collaborative virtual environment technologies. He gained his PhD in 1989 and has been involved in a number of collaborative projects with BT since then including COSMOS, Virtuosi, Inhabiting the Web and now Network Architectures for Inhabited Television.